

Signal Processing

Exercise 1 – Fourier Series – Solution

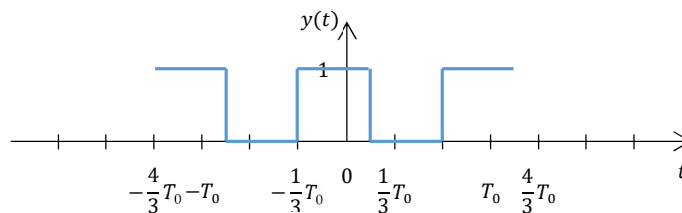
Summer Semester 2026

Question 1

The periodic signal $f(t)$ is defined as

$$f(t) = \begin{cases} 1, & \text{für } -\frac{T_0}{3} \leq t < +\frac{T_0}{6} \\ 0, & \text{für } +\frac{T_0}{6} \leq t < +\frac{2}{3}T_0 \end{cases} ; \quad f(t) = f(t + n \cdot T_0), \quad n = \pm 1, \pm 2, \dots$$

a) Sketch the signal $f(t)$ for $t = -\frac{4}{3}T_0$ to $t = \frac{7}{6}T_0$.



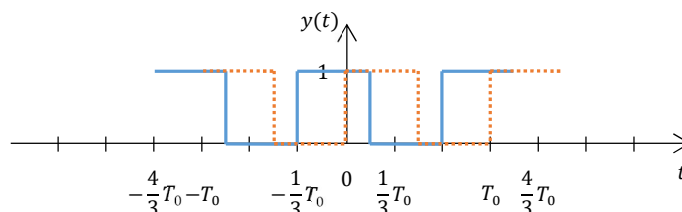
b) Calculate the real and complex Fourier coefficients.

1. Coefficient a_0 of the DC component:

$$a_0 = \frac{2}{T_0} \int_{-\frac{1}{3}T_0}^{\frac{1}{6}T_0} 1 \, dt = \frac{2}{T_0} [t]_{-\frac{1}{3}T_0}^{\frac{1}{6}T_0} = 1$$

2. Determination of a_n and b_n :

Instead of solving two integrals, shift $f\left(t - \frac{T_0}{3}\right) = g(t)$.



We get a point symmetric signal (odd) with respect to the DC offset.

According to definition: $a_{gn} = 0 \rightarrow$ Calculation of b_{gn}

With $g(t) = 1$ in $t = \left\{0; \frac{T_0}{2}\right\}$

$$b_{gn} = \frac{2}{T_0} \int_0^{\frac{T_0}{2}} g(t) \cdot \sin\left(\frac{2\pi n t}{T_0}\right) dt = \frac{2}{T_0} \left[-\frac{T_0}{2\pi n} \cdot \cos\left(\frac{2\pi n t}{T_0}\right) \right]_0^{\frac{T_0}{2}}$$

$$b_{gn} = \frac{1}{\pi n} [1 - \cos(\pi n)]$$

$$b_{gn} = 0 \quad \text{for } n = 2, 4, 6 \dots$$

$$b_n = \frac{2}{\pi n} \quad \text{for } n = 1, 3, 5 \dots$$

3. Calculation of c_{gn} :

$$c_{gn} = \frac{1}{2} (a_{gn} - j b_{gn})$$

$$c_{gn} = 0 \quad \text{for } n = 2, 4, 6 \dots$$

$$c_{gn} = \frac{-j}{\pi n} \quad \text{for } n = 1, 3, 5 \dots$$

4. Consider time shift: $g\left(t + \frac{T_0}{3}\right) = f(t)$

$$x(t - \tau) \quad \text{O} \text{---} \bullet \quad X(f) \cdot e^{-j2\pi f \tau}$$

$$c_n = c_{gn} \cdot e^{j2\pi n \cdot \frac{1}{T_0} \cdot \frac{T_0}{3}} \quad n \in \mathbb{Z}$$

5. Calculate c_n :

$$c_n = 0 \quad \text{for } n = 2, 4, 6 \dots$$

$$c_n = \frac{-j}{\pi n} \cdot e^{j\frac{2\pi}{3}n} \quad \text{for } n = 1, 3, 5 \dots$$

6. Calculate a_n and b_n :

$$a_n = c_n + c_{-n} \quad b_n = j(c_n - c_{-n})$$

$$a_n = 0 \quad \text{for } n = 2, 4, 6 \dots$$

$$a_n = -\frac{j}{\pi n} \left(e^{j\frac{2\pi}{3}n} - e^{-j\frac{2\pi}{3}n} \right) = -\frac{j}{\pi n} \left(e^{j\frac{2\pi}{3}n} - e^{-j\frac{2\pi}{3}n} \right) \cdot \frac{2j}{2j} = \frac{2}{\pi n} \cdot \sin\left(\frac{2\pi}{3} \cdot n\right)$$

$$\text{for } n = 1, 3, 5 \dots$$

$$b_n = 0 \quad \text{for } n = 2, 4, 6 \dots$$

$$b_n = \frac{1}{\pi \cdot n} \left(e^{j\frac{2\pi}{3}n} + e^{-j\frac{2\pi}{3}n} \right) = \frac{2}{\pi \cdot n} \cdot \cos \left(\frac{2\pi}{3} \cdot n \right) \quad \text{for } n = 1, 3, 5 \dots$$

c) Sketch the line spectrum (magnitude spectrum) of $|c_n|$ for $n \leq 5$.

DC component $c_0 = 0,5$

$$c_1 = 0,28 + 0,16j \quad c_{-1} = c_1^* = 0,28 - 0,16j \quad |c_{1,-1}| = 0,32$$

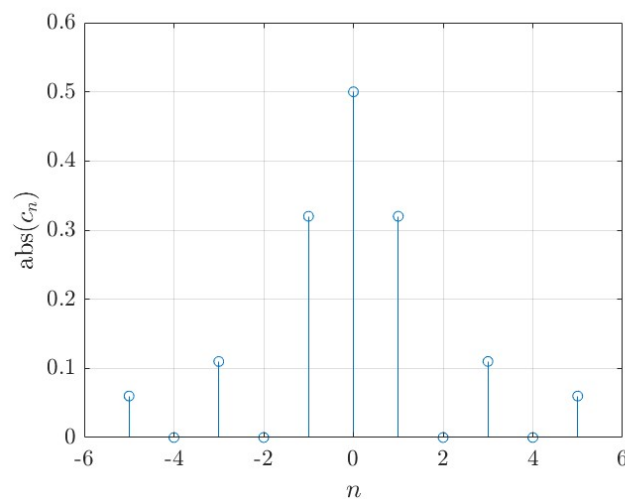
$$c_2 = 0 \quad c_{-2} = c_2^* = 0 \quad |c_{2,-2}| = 0$$

$$c_3 = 0,11j \quad c_{-3} = c_3^* = -0,11j \quad |c_{3,-3}| = 0,11$$

$$c_4 = 0 \quad c_{-4} = c_4^* = 0 \quad |c_{4,-4}| = 0$$

$$c_5 = -0,06 + 0,03j \quad c_{-5} = c_5^* = -0,06 - 0,03j \quad |c_{5,-5}| = 0,06$$

Magnitude spectrum:



One can determine the **amplitude A_n of the n -th signal component!**

Attention: $A_n = 2 \cdot |c_n|$ for $n > 0$

Question 2

We will consider the signal $y(t) = \cos\left(\omega_0 \cdot \left(t - \frac{T_0}{3}\right)\right)$. Calculate the real and complex Fourier coefficients.

$$y(t) = \cos(\omega_0 \cdot t) = \frac{1}{2}(e^{j\omega_0 t} + e^{-j\omega_0 t}) = \frac{1}{2}e^{-j\omega_0 t} + \frac{1}{2}e^{j\omega_0 t}$$

Complex Fourier series: $y(t) = \sum_{-\infty}^{\infty} c_n \cdot e^{j\omega_0 n t}$

→ Comparison of coefficients for $n = -1; 1 \rightarrow c_1 = c_{-1} = \frac{1}{2}$

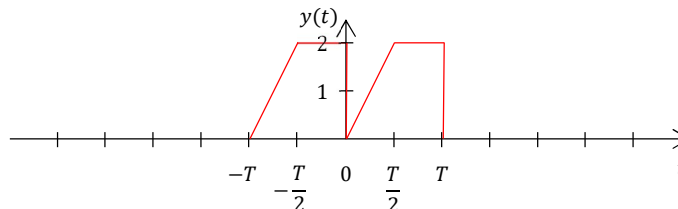
Consider time shift $a = -\frac{T_0}{3} \rightarrow$ Cheat sheet $\rightarrow$ Property Fourier transform $\rightarrow$ Time domain shift equals phase shift in frequency domain, i.e., multiplication of c_1 with $e^{j\omega a}$

$$c_1 = \frac{1}{2} \cdot e^{-j\frac{2}{3}\pi} = -0,25 - 0,433j ; c_{-1} = \frac{1}{2} \cdot e^{j\frac{2}{3}\pi} = -0,25 + 0,433j ;$$

$$a_1 = -0,25 - 0,433j + (-0,25 + 0,433j) ; b_1 = j((-0,25 - 0,433j) - (-0,25 + 0,433j))$$

Question 3

The periodic signal $y(t)$ is given by the following sketch.



a) Calculate the coefficient a_0 of the DC component.

$$a_0 = \frac{2}{T} \int_0^T y(t) dt = \frac{2}{T} \int_0^{\frac{T}{2}} m \cdot t dt + \frac{2}{T} \int_{\frac{T}{2}}^T K dt = \frac{2}{T} \int_0^{\frac{T}{2}} \frac{4}{T} \cdot t dt + \frac{2}{T} \int_{\frac{T}{2}}^T 2 dt$$

$$a_0 = \frac{8}{T^2} \left[-\frac{1}{2} t^2 \right]_0^{\frac{T_0}{2}} + \frac{4}{T} [t]_{\frac{T}{2}}^T = 3$$

- b) Derive a formula which allows the calculation of each coefficient a_n of the even signal component.

$$a_n = \frac{2}{T} \int_0^T y(t) \cdot \cos(\omega_0 n t) dt = \frac{8}{T^2} \int_0^{\frac{T}{2}} t \cdot \cos(\omega_0 n t) dt + \frac{4}{T} \int_{\frac{T}{2}}^T \cos(\omega_0 n t) dt$$

→ Integration by parts: $\int_a^b v'(x) \cdot u(x) dx = [v(x) \cdot u(x)]_a^b - \int_a^b v(x) \cdot u'(x) dx$
 $u(t) = t \quad v'(t) = \cos(\omega_0 n t)$

$$a_n = \frac{4}{T} \cdot \frac{1}{\omega_0 n} [\sin(\omega_0 n t)]_{\frac{T}{2}}^T + \frac{8}{T^2} \left[\frac{1}{\omega_0 n} [t \cdot \sin(\omega_0 n t)]_0^{\frac{T}{2}} - \frac{1}{\omega_0 n} \int_0^{\frac{T}{2}} \sin(\omega_0 n t) dt \right]$$

With: $\omega_0 = \frac{2\pi}{T}$

$$a_n = 0 + \frac{8}{T^2} \left[\frac{1}{\omega_0 n} \left[\frac{T}{2} \cdot \sin\left(\frac{2\pi}{T} n \frac{T}{2}\right) \right] - \frac{1}{\omega_0 n} \cdot \frac{1}{\omega_0 n} \cdot (-1) \cdot [\cos(\omega_0 n t)]_0^{\frac{T}{2}} \right]$$

$$a_n = \frac{8}{T^2} \left[\frac{1}{\omega_0^2 n^2} \left[\cos\left(\frac{2\pi}{T} n \frac{T}{2}\right) - \cos(0) \right] \right]$$

$$a_n = \frac{2}{\pi^2 n^2} [(-1)^n - 1]$$

$$a_n = \frac{-4}{\pi^2 n^2} \quad \text{for } n = 1, 3, 5 \dots$$

$$a_n = 0 \quad \text{for } n = 2, 4, 6 \dots$$

- c) Derive a formula which allows the calculation of each coefficient b_n of the odd signal component.

$$b_n = \frac{2}{T} \int_0^T y(t) \cdot \sin(\omega_0 n t) dt = \frac{8}{T^2} \int_0^{\frac{T}{2}} t \cdot \sin(\omega_0 n t) dt + \frac{4}{T} \int_{\frac{T}{2}}^T \sin(\omega_0 n t) dt$$

$$u(t) = t \quad v'(t) = \sin(\omega_0 n t)$$

$$b_n = \frac{-4}{T} \cdot \frac{1}{\omega_0 n} [\cos(\omega_0 n t)]_{\frac{T}{2}}^T + \frac{8}{T^2} \left[\frac{-1}{\omega_0 n} [t \cdot \cos(\omega_0 n t)]_0^{\frac{T}{2}} - \frac{-1}{\omega_0 n} \int_0^{\frac{T}{2}} \cos(\omega_0 n t) dt \right]$$

$$b_n = \frac{-4}{T \omega_0 n} [1 - (-1)^n] + \frac{8}{T^2} \left[\frac{-1}{\omega_0 n} [t \cdot \cos(\omega_0 n t)]_0^{\frac{T}{2}} + \frac{1}{\omega_0^2 n^2} [\sin(\omega_0 n t)]_0^{\frac{T}{2}} \right]$$

$$b_n = \frac{-4}{T \omega_0 n} [1 - (-1)^n] + \frac{8}{T^2} \left[\frac{-T^2}{4\pi n} (-1)^n + \frac{T^2}{4\pi n} \sin(\pi n) \right]$$

$$b_n = \frac{-2}{\pi n} + \frac{2}{\pi n} (-1)^n - \frac{2}{\pi n} (-1)^n = -\frac{2}{\pi n} \quad \text{for } n = 1, 2, 3, 4, 5 \dots$$

Exercise 4

Let $y(t)$ be a periodic signal. Describe how the Fourier coefficients of the following variations $y_{\text{var}}(t)$ can be derived from the Fourier coefficients of $y(t)$.

a) $y_{\text{var}}(t) = y(t) + 4$

A constant is added to the function, therefore, only the offset component is varied:

$$a_{0,\text{var}} = a_0 + 8$$

or

$$c_{0,\text{var}} = c_0 + 4$$

a) $y_{\text{var}}(t) = 2 \cdot y(t)$

The function is scaled with the factor 2, which results in a scaling of the Fourier coefficients by the same factor:

$$a_{0,\text{var}} = 2 \cdot a_0$$

$$a_{n,\text{var}} = 2 \cdot a_n$$

$$b_{n,\text{var}} = 2 \cdot b_n$$

or

$$c_{n,\text{var}} = 2 \cdot c_n$$

b) $y_{\text{var}}(t) = y(3t)$

The function is compressed in time; therefore, only T_0 changes, but not the coefficients.